

- Phylogenies based in part on rRNA genes suggest that eukaryotes are most closely related to archaea, while data from some other genes suggest a closer relationship to bacteria.
- Genetic analyses indicate that extensive **horizontal gene transfer** has occurred throughout the evolutionary history of life.

7. Why was the five-kingdom system abandoned for a three-domain system?

TEST YOUR UNDERSTANDING

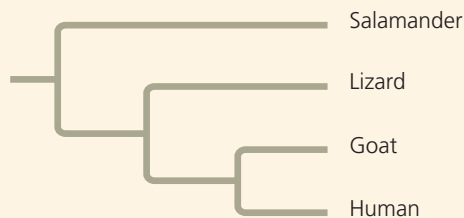
➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- In a comparison of birds and mammals, the condition of having four limbs is
 - a shared ancestral character.
 - a shared derived character.
 - a character useful for distinguishing birds from mammals.
 - an example of analogy rather than homology.
- To apply parsimony to constructing a phylogenetic tree,
 - choose the tree that assumes all evolutionary changes are equally probable.
 - choose the tree in which the branch points are based on as many shared derived characters as possible.
 - choose the tree that represents the fewest evolutionary changes, in either DNA sequences or morphology.
 - choose the tree with the fewest branch points.

Levels 3-4: Applying/Analyzing

- VISUAL SKILLS** In Figure 26.4, which similarly inclusive taxon is represented as descending from the same common ancestor as Canidae?
 - Felidae
 - Mustelidae
 - Carnivora
 - Lutra*
- Three living species X, Y, and Z share a common ancestor T, as do extinct species U and V. A grouping that consists of species T, X, Y, and Z (but not U or V) makes up
 - a monophyletic taxon.
 - an ingroup, with species U as the outgroup.
 - a paraphyletic group.
 - a polyphyletic group.
- VISUAL SKILLS** Based on the tree below, which statement is correct?



- Lizards and goats form a sister group.
 - Salamanders are a sister group to the group containing lizards, goats, and humans.
 - Salamanders are more closely related to lizards than to humans.
 - Goats and humans are the only sister group shown in this tree.
- If you were using cladistics to build a phylogenetic tree of cats, which of the following would be the best outgroup?
 - wolf
 - domestic cat
 - lion
 - leopard

- VISUAL SKILLS** The relative lengths of the frog and mouse branches in the phylogenetic tree in Figure 26.13 indicate that
 - frogs evolved before mice.
 - mice evolved before frogs.
 - the homolog evolved more rapidly in the mouse lineage.
 - the homolog evolved more slowly in the mouse lineage.

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** Darwin suggested looking at a species' close relatives to learn what its ancestors may have been like. Explain how his suggestion anticipates recent methods, such as phylogenetic bracketing and the use of outgroups in cladistic analysis.
- SCIENTIFIC INQUIRY • DRAW IT** (a) Draw a phylogenetic tree based on the characters in the table below. Draw hatch marks on the tree to indicate the origin of characters 1–6. (b) Assume that tuna and dolphins are sister species and redraw the phylogenetic tree accordingly. Use hatch marks to indicate the origin of characters 1–6. (c) Determine how many evolutionary changes are required in each tree. Identify the most parsimonious tree.

Character	Lancelet (outgroup)	Lamprey	Tuna	Salamander	Turtle	Leopard	Dolphin
(1) Backbone	0	1	1	1	1	1	1
(2) Hinged jaw	0	0	1	1	1	1	1
(3) Four limbs	0	0	0	1	1	1	1*
(4) Amnion	0	0	0	0	1	1	1
(5) Milk	0	0	0	0	0	1	1
(6) Dorsal fin	0	0	1	0	0	0	1

*Although adult dolphins have only two obvious limbs (their flippers), as embryos they have two hind-limb buds, for a total of four limbs.

- WRITE ABOUT A THEME: INFORMATION** In a short essay (100–150 words), explain how genetic information—along with an understanding of the process of descent with modification—enables scientists to reconstruct phylogenies that extend hundreds of millions of years back in time.
- SYNTHESIZE YOUR KNOWLEDGE**



This West Indian manatee (*Trichechus manatus*) is an aquatic mammal. Like amphibians and reptiles, mammals are tetrapods (vertebrates with four limbs). Explain why manatees are considered tetrapods even though they lack hind limbs, and suggest traits that manatees likely share with leopards and other mammals (see Figure 26.12b). Discuss how early members of the manatee lineage might have differed from today's manatees.

For selected answers, see Appendix A.